

Problem Set 2

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Problem 1

The evolution of x_t is described by a linear system

$$x_{t+1} = Ax_t, \quad A = \begin{pmatrix} 1 & -0.5 \\ -1 & 1.5 \end{pmatrix}.$$

1. Find the steady state of the system. Is it a source, a sink, or a saddle point?
2. Draw the iso-lines and show with arrows the directions of the trajectories.
3. Decompose matrix A and show the eigenvectors on the phase diagram.
4. Consider $x_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, compute y_0 of the corresponding decoupled system, and show dynamics for periods $t = 1$ and $t = 2$ on the phase diagram.

Problem 2

Consider a household with a quadratic utility

$$\int_0^\infty e^{-\rho t} \left(\alpha C_t - \frac{1}{2} C_t^2 \right) dt,$$

which has a “bliss point” that maximizes static utility $C^* = \alpha > 0$. Note that consumption is allowed to be negative in this case. The household earns constant labor income $Y_t = Y$ and is subject to a standard budget constraint

$$\dot{B}_t = rB_t + Y - C_t.$$

1. Write down the Hamiltonian and derive the optimality conditions.
2. Obtain the Euler equation and the transversality condition.
3. What is the dynamic system that describes the evolution of C_t and B_t ? How many steady states does it have?
4. Draw the phase diagram and show the optimal consumption path. Consider separately two cases: (i) $r > \rho$ and (ii) $r < \rho$.

Problem 3

Consider the investment problem from Problem Set 1: given initial value of capital K_0 , a firm chooses investment path $\{I_t\}_{t=0}^{\infty}$ to maximize profits

$$\sum_{t=0}^{\infty} \beta^t \left[AK_t^\alpha - I_t - \frac{\phi}{2} \frac{I_t^2}{K_t} \right]$$

subject to the capital law of motion

$$K_{t+1} = (1 - \delta)K_t + I_t.$$

Assume that $\delta = 0$ and define $Q_t \equiv \beta^{-t} \lambda_t$, where λ_t is the Lagrange multiplier from the optimization problem.

1. Write down the (non-linear) system of equations for K_t and Q_t .
2. Determine the state variables and the control variables in this system.
3. Solve for the steady state.
4. Log-linearize the equations for K_t and Q_t around the steady state. Denote the log deviations with small letters k_t and q_t .
5. Is the Blanchard-Kahn condition for a unique solution satisfied?
6. Solve for the saddle path using the Blanchard-Kahn method.
7. Solve for the optimal dynamics of k_t .
8. Draw a phase diagram.